Predicting a Water Quality Index Without Data Leakage

A study using classical machine learning

Project 18 · Gradient Boosting and Deep Neural Networks · Phase 1 (AML)

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Abstract. A water quality index turns many water measurements into one score between 0 and 100. In this project we train models to predict this score (the CCME index) from a small set of readings, and we compare four models: a baseline that always guesses the average, Ridge Regression, Random Forest and XGBoost. The index is calculated from eight measurements, so giving all eight to a model lets it simply work out the formula ($R^2 = 0.991$), which proves nothing. We therefore hide five of these measurements from the models. We test on monitoring stations the models have never seen, and separately on later years. On the unseen stations, XGBoost was best with MAE 7.543, RMSE 11.058 and $R^2 = 0.293$, slightly ahead of Random Forest ($R^2 = 0.278$). On later years, XGBoost reached $R^2 = 0.243$. The models predict polluted water to be better than it is, and much of their skill comes from knowing the country and the past readings of a station. Overall, tree models work better than a straight-line model, but a lot of the variation in the score is still unexplained.

1. Introduction

Authorities test rivers, lakes and coastal water to protect people and nature. A water quality index combines many measurements into one number, which is easier to report and compare. If a model could predict this number from a few simple readings, it could help with screening. But a high score from a model is only meaningful if the test is fair. In particular, if the model is given the same measurements that the index is calculated from, it can just recompute the formula instead of really predicting anything.

In this project we predict `target_ccme_value`, a CCME water quality index between 0 and 100 (higher means cleaner water). Our main contributions are:

- an experiment that shows how much the formula leaks when all measurements are given, and a smaller input set that avoids it;
- a fair test on stations the models have never seen, plus a second test on later years;
- a comparison of four models using MAE, RMSE and R^2, with confidence intervals and a look at where the models go wrong.

The report first reviews related work, then describes the data and methods, and finally gives the results, limitations and our plan for Phase 2.

2. Related work

The CCME index is based on how many measurements break their guideline, how often, and by how much [1]. Reviews of surface-water quality describe the many water quality indices and pollution indices in use [2]. They also point out weaknesses: one very bad measurement can be hidden by good ones, information about how measurements interact is lost, and the choice of measurements is partly subjective [3]. Karim et al. released a large dataset of surface-water measurements from 1940 to 2023 [4]. We use only a cleaned part of it, so our counts are different from the original dataset.

Random Forests use many decision trees built on random parts of the data [5]. Gradient boosting builds a model step by step, where each step fixes the mistakes of the earlier ones [6], and XGBoost is a fast and well-controlled version of this idea [7]. SHAP is a method that shows how much each input pushed a prediction up or down [8]; it shows what the model uses, not what actually causes pollution.

Other studies found that deep neural networks do not always beat tree models on table-like data [9, 10].

Water quality studies use different tasks and measures, so their results are hard to compare. Ooko et al. found that Random Forest worked best, with 79% accuracy, in a water quality classification study [11]. Rahman et al. reported 97.33% accuracy for sorting samples into five CCME classes on the Canadian part of the same dataset, using the eight measurements as inputs [12]. Since the index is calculated from those eight measurements, such scores cannot be compared with ours. Section 4.1 shows that giving a model all eight measurements gives $R^2 = 0.991$.

The gap we focus on is the way models are tested. Scores can look better than they really are when the model is given the measurements the target is made from, or when the same station appears in both training and test data. We hide five of the main measurements, test on stations the model has not seen, add a test on later years, and use the same measures for every model. We do not claim that earlier studies all made these mistakes.

3. Dataset and target

After cleaning, our dataset has **259,774 rows** from **9,940 monitoring stations** in England, Ireland, the USA and Canada, from 2000 to 2023. The target is the CCME score. Many samples are exactly 100 (43.7% of the rows), so the scores pile up at the top. The original dataset has 2.82 million records from five countries [4]; we keep a part of it that suits our fair-testing plan.

We removed effluent, sewage and drainage samples (they are not surface water), records before 2000, exact duplicates, and impossible values. We also found that the authors of the source data had filled many gaps with one repeated number for each country and water type (for example, dissolved oxygen 10.2 in 360,401 English rows). Real measurements do not repeat like this, so we found these values by how often they appeared and marked them as missing. They may still affect the score, because the index was calculated after the gaps were filled. Finally, we kept whole stations when reducing the size of the dataset, so that each station keeps its full history.

Cleaning step	Rows
Effluent, sewage, drainage removed	635,532
Records before 2000 removed	243,089
Exact duplicates removed	7,277
Rows with fewer than 5 of 8 real measurements removed	474,227
Final dataset after keeping whole stations	259,774

4. Method

4.1 Avoiding data leakage

The index is calculated from eight measurements: ammonia, BOD, dissolved oxygen, orthophosphate, pH, temperature, nitrogen and nitrate. To see how much a model can recover, we trained XGBoost on different sets of inputs and scored it on the validation data. With all eight measurements the score was $R^2 = 0.991$, and with our first proposed set it was 0.982. This means the model had simply learned the formula. So we removed ammonia, BOD, orthophosphate, nitrogen and nitrate from the inputs. Dissolved oxygen, pH and temperature are also part of the index, so some dependence remains, but each of them alone explains very little (R^2 at most 0.04).

Inputs given to XGBoost	Validation R^2
All eight measurements	0.991
First proposal (pH, temperature, ammonia, orthophosphate, nitrogen, type)	0.982
Dissolved oxygen, pH, temperature only	0.127
+ water type, country, month	0.268
+ extra features and station history (chosen)	0.343
Baseline (always the average)	−0.005

4.2 Features and preprocessing

The models use 14 inputs: dissolved oxygen, pH, temperature, the month (as a sine and a cosine, so that December is next to January), oxygen saturation, dissolved oxygen × temperature, the station's average past dissolved oxygen, pH and temperature, the number of earlier samples at the station, days since the previous sample, water body type and country. The past averages only use earlier sampling dates of the same station, so nothing from the future is used. Missing values are filled with the median of the training data (and a "was missing" flag is added), categories are one-hot encoded, and values are scaled for Ridge only. All of this is learned from the training data only.

4.3 Train, validation and test split

Our main split is **by station**. All rows of a station go to only one part: 70% of rows for training, 10% for validation and 20% for testing (181,747 / 26,054 / 51,973 rows from 6,583 / 1,136 / 2,221 stations). This way the test shows how a model does at a place it has never seen. A random split of rows would put the same station in both parts, and the model could do well just by remembering the station. Our second split is **by year**: training up to 2015, validation 2016–2017 and test 2018 onwards. It checks how the models do on later data. We choose settings using the validation data only, and use the test data once at the end.

4.4 Models and tuning

The baseline always predicts the average of the training data. Ridge Regression is a straight-line model with a penalty that keeps the numbers small. Random Forest averages the answers of many deep decision trees, which makes it more stable. XGBoost builds trees one after another, where each new tree tries to fix the mistakes of the earlier ones. We tuned each model on the validation data by trying a small grid of settings, changing one setting at a time: Ridge penalty α from 0.001 to 1000; Random Forest depth 6–16 and minimum leaf size 2–20 (300 trees); XGBoost depth 4–10, learning rate 0.03–0.1 and 300 or 600 trees.

The settings with the lowest validation RMSE were chosen: Ridge $\alpha = 0.001$; Random Forest depth 16 and minimum leaf size 5; XGBoost depth 8, learning rate 0.03, 300 trees, row sampling 0.8, feature sampling 0.8 and L2 weight 1.0. The random seed was 42.

4.5 Metrics and confidence intervals

$$\textbf{MAE} = (1/n) \cdot \Sigma\, |y_i - \hat{y}_i|$$

$$\textbf{RMSE} = \sqrt{[\ (1/n) \cdot \Sigma\, (y_i - \hat{y}_i)^2\ ]}$$

$$\textbf{R}^2 = 1 - \Sigma\, (y_i - \hat{y}_i)^2 \,/\, \Sigma\, (y_i - \bar{y})^2$$

MAE is the average size of the error, and RMSE is similar but gives more weight to big errors. For both, lower is better. R^2 tells us how much better the model is than always guessing the average: 1 is perfect and 0 is no better than the average. A test set is only one sample of stations, so a score has some luck in it. To measure this, we repeated the evaluation 2,000 times on stations drawn again at random (keeping all rows of a station together, because the rows of one station are similar) and used the middle 95% of the results as the 95% confidence interval. To compare two models fairly, we used the

same re-drawn stations for both and looked at the difference between their errors.

5. Results

5.1 Validation results

On the validation data XGBoost was best (MAE 7.729, RMSE 11.205, $R^2 = 0.348$), just ahead of Random Forest (7.773, 11.247, 0.343). Ridge reached $R^2 = 0.219$ and the baseline −0.005. On the training data, Random Forest scored $R^2 = 0.558$ and XGBoost 0.461, so Random Forest fits the training data more closely and overfits more. Ridge scored about the same on training data (0.211), because it is too simple to learn much.

5.2 Test on unseen stations

Model	MAE	RMSE	R^2	95% CI for R^2
Baseline (average)	9.570	13.159	−0.001	[−0.009, 0.000]
Ridge	8.329	11.733	0.205	[0.174, 0.233]
Random Forest	7.598	11.179	0.278	[0.230, 0.324]
XGBoost	**7.543**	**11.058**	**0.293**	[0.246, 0.340]

XGBoost is the best model. The confidence intervals of XGBoost and Random Forest overlap, but both models are affected by the same lucky or unlucky stations. So we compared them on the same re-drawn stations: XGBoost's RMSE was lower by 0.121 (95% interval −0.186 to −0.059). This means XGBoost is better in a consistent way, but only by a small amount. Both tree models are clearly better than Ridge (XGBoost's RMSE is 0.674 lower) and than the baseline (2.101 lower). Still, R^2 below 0.30 means that most of the variation in the score is not explained.

The test R^2 is lower than the validation R^2 (0.293 against 0.348 for XGBoost). One reason is that we picked the settings that looked best on the validation data. Another is probably that the test stations are different and harder. We did not separate these two reasons.

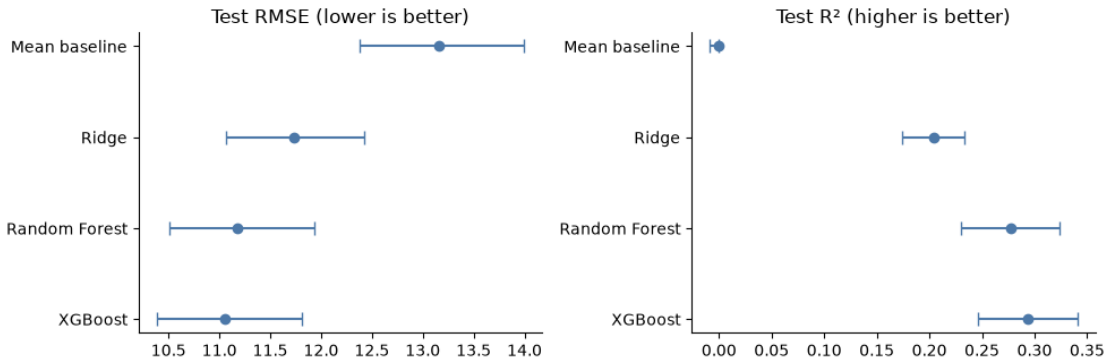


Figure 1. Confidence intervals of the test scores.

5.3 Test on later years

For this test the models were trained again on the year split, with the same settings. These numbers should not be mixed with the station test, because the test data are different. For example, the baseline gets a negative R^2 here, which shows that the scores changed after 2015. The order of the models stays the same: XGBoost, Random Forest, Ridge, baseline.

Model (2018 onwards)	MAE	RMSE	R^2
Baseline (average)	9.101	10.693	−0.338
Ridge	6.738	8.786	0.097
Random Forest	5.596	8.310	0.192
XGBoost	**5.449**	**8.047**	**0.243**

5.4 Where do the models go wrong?

The overall score hides big differences between countries. England makes up 70% of the test rows, and the model helps least there (its MAE is only 6% lower than the baseline's). Ireland shows the biggest improvement, but 82% of Irish samples are exactly 100, so there is little left to explain and its R^2 is close to zero. The USA and Canada have few rows, so their results are not very reliable.

Country	Rows	MAE	Baseline MAE	R^2
England	36,396	9.25	9.86	0.194
Ireland	14,443	3.30	8.91	0.026
USA	580	5.88	9.35	0.156
Canada	554	7.63	8.15	0.080

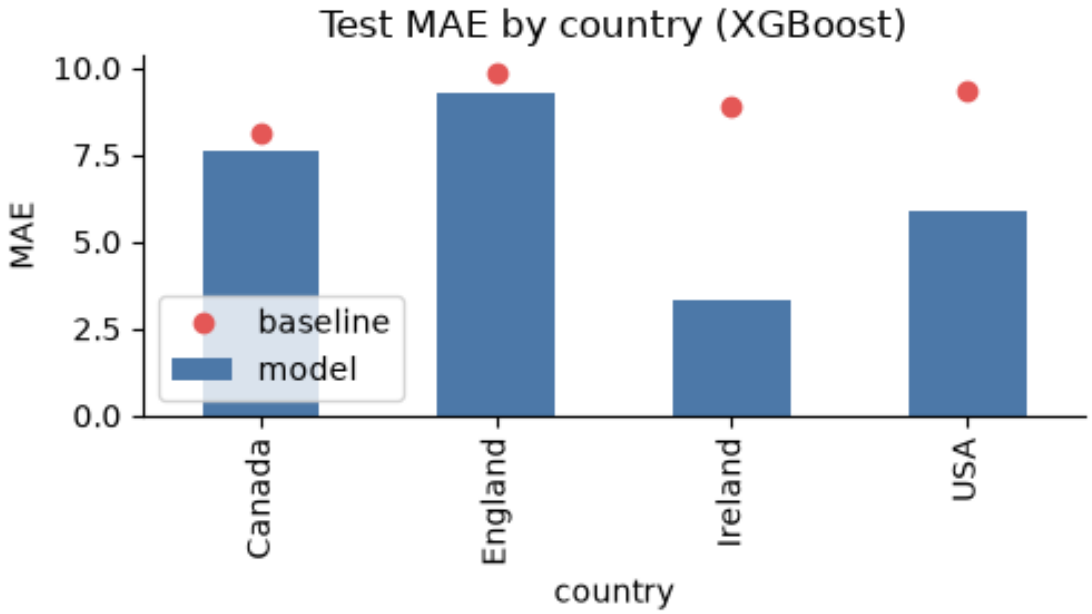


Figure 2. XGBoost test performance by country.

The error also depends on how good the water really is. When the true score is below 60 (poor water), the model guesses about 31 points too high on average, and for perfect water (100) it guesses about 5.5 points too low. So the guesses are pulled towards the middle. This happens because the five main measurements are hidden, so the model cannot tell when water is really bad, and 44% of all samples are exactly 100, so a “fairly good” guess is usually safe. Several of the hardest stations are poor-quality places near sewage works or industrial discharges that the model predicts as good.

True score	Rows	MAE	Mean error
below 60	2,719	30.98	+30.98
60 to 80	6,032	12.52	+12.15
80 to 95	20,502	5.19	−1.90
100	22,720	5.54	−5.52

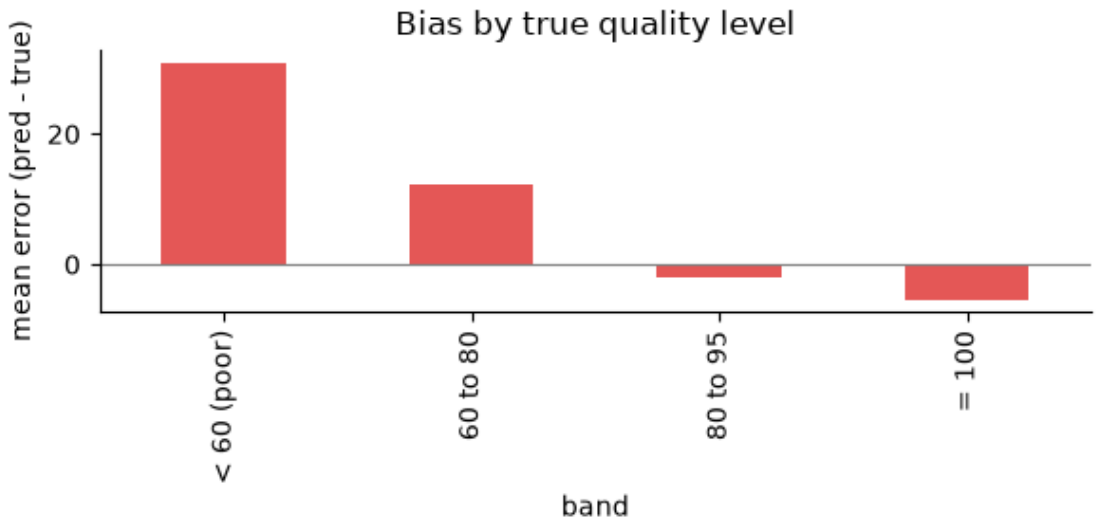


Figure 3. Average error of XGBoost by true water quality level.

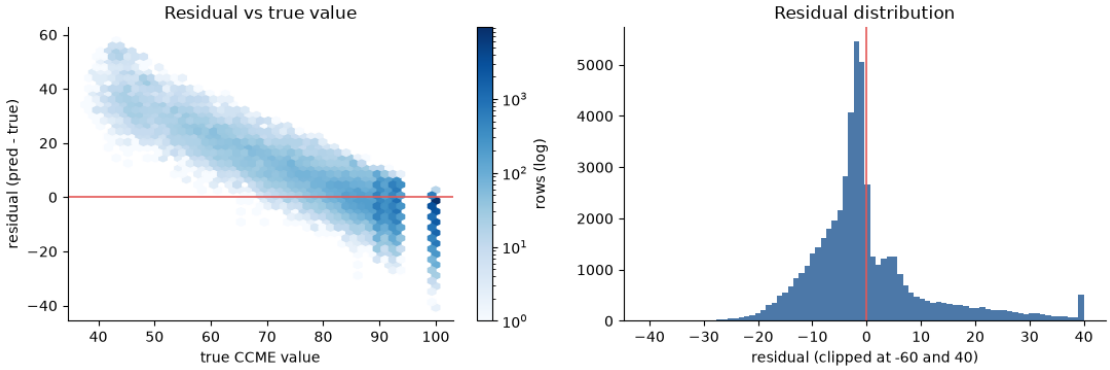


Figure 4. Errors of the test predictions.

5.5 Which inputs matter?

To see how much the models depend on some inputs, we trained XGBoost again with them removed and checked the validation score. With all inputs, R^2 was 0.348. Without country it dropped to 0.274, without the station history to 0.271, and without both to 0.162. SHAP gives the same picture on 3,000 test rows: country is the most important input (average effect 4.75 points), followed by the station's past dissolved oxygen (1.40), temperature (1.32) and the station's past pH (1.19). So a good part of the score comes from knowing which country and station a sample is from. This may not work well in a new region. SHAP shows what the model uses, not what causes the water quality.

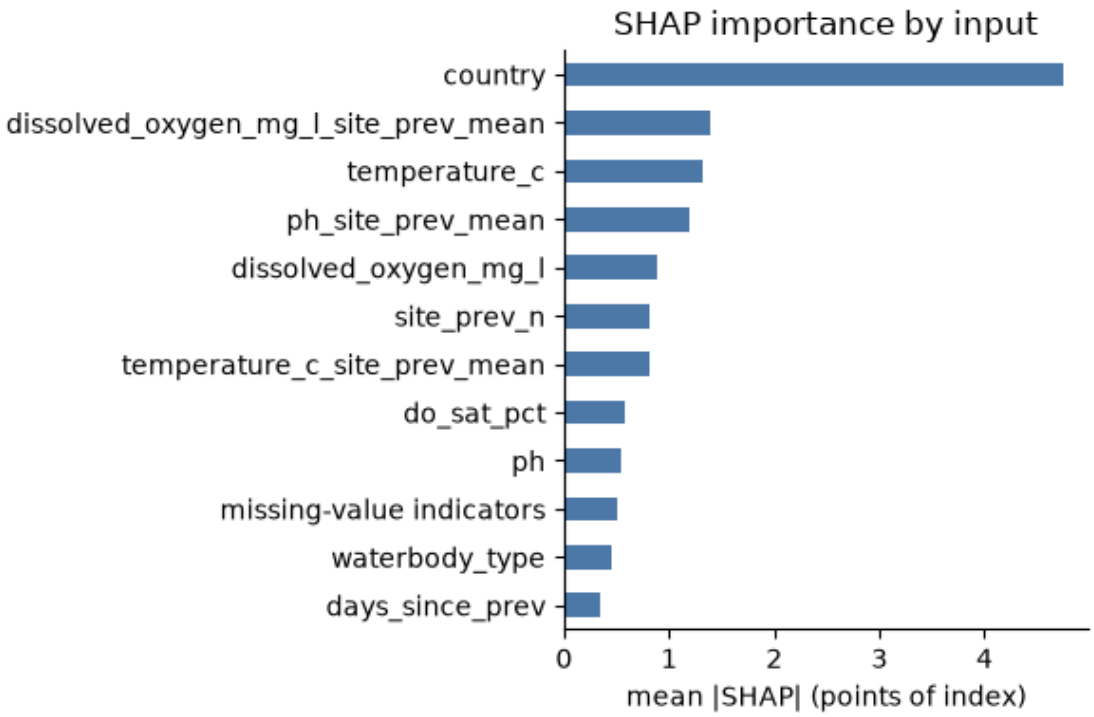


Figure 5. Importance of each input for XGBoost (SHAP).

6. Discussion

The tree models work better than the baseline and Ridge, even with the reduced set of inputs. XGBoost is best on both test types, and Random Forest is close; the comparison on the same re-drawn stations shows that XGBoost's lead is real but small. On later years the errors are lower but R^2 is also lower than on new stations. This is because the data are different, so we do not treat it as one split being better than the other. Many samples sit at 100, so one measure alone does not tell the whole story, and we report MAE, RMSE and R^2 together with results by country and by quality level. The biggest weakness is that polluted water is predicted as better than it is, which is the more harmful mistake in real monitoring.

7. Limitations and future work

The model cannot replace laboratory tests or official checks. Countries and monitoring practices differ, and the filled-in values in the source data may affect the score. The test R^2 is moderate, which is expected because the five main measurements are hidden on purpose. We chose settings using one validation split and did not use cross-validation, so the validation scores are somewhat optimistic. The best Random Forest depth and the best XGBoost learning rate were at the edge of what we tried, so slightly better settings may exist. England has most of the data, while the USA and Canada have very little.

In Phase 2 we plan to train a small neural network (MLP) on the same inputs and the same splits, and compare it fairly without touching the test set while choosing the model. We also want to check how the models work on stations from regions with little data.

8. Conclusion

We compared an average-only baseline, Ridge Regression, Random Forest and XGBoost for predicting the CCME water quality index without using the measurements it is calculated from. On unseen stations XGBoost gave MAE 7.543, RMSE 11.058 and $R^2 = 0.293$, slightly but consistently better than Random Forest. On later years it gave MAE 5.449, RMSE 8.047 and $R^2 = 0.243$. Tree models are good starting points, but they do not predict the index very accurately: they over-estimate poor water, and much of their skill comes from the country and the station history. The next step is a fair comparison with a neural network.

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